

Additional analytic benchmarks: complete quantitative results

Case	Grid	Role	Velocity L2 (%)	Pressure metric (%)	Shear L2 (%)	Pass
abc	24x24x24	spatial	0.005699	1.69278	-	True
abc	32x32x32	spatial	0.003209	0.954533	-	True
abc	48x48x48	spatial	0.001427	0.42498	-	True
advected-shear	16x16x4	spatial	0.620363	0	0.895360	True
advected-shear	24x24x4	spatial	0.276903	0	0.399650	True
advected-shear	32x32x4	spatial	0.156024	0	0.225186	True
abc	16x16x16	negative-control	0.012786	3.78203	-	False
advected-shear-lbm	32x32	comparison	0.230176	1.42962e-10	0.332209	True
advected-shear-lbm	64x64	comparison	0.054381	1.78702e-10	0.078488	True

ABC pressure: relative L2 at final midpoint. Shear pressure: max absolute pressure / initial dynamic pressure.

All declared physical errors <3% on formal grids; n16 ABC is an intentional pressure failure control.

Shear component error is checked separately to avoid dilution by constant mean flow.

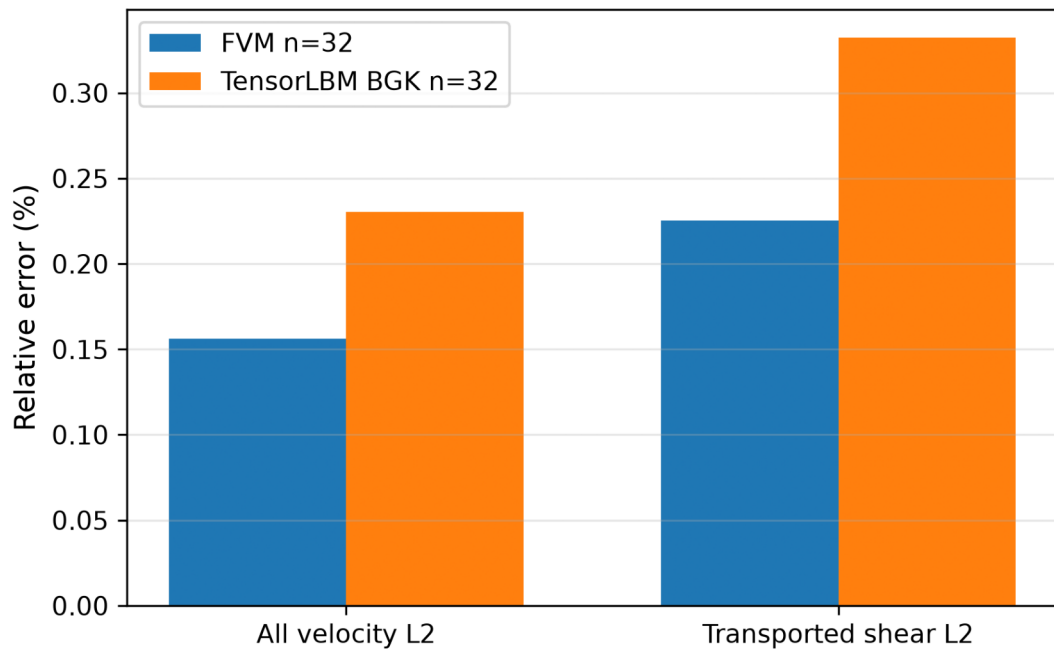
ABC is a genuine 3D periodic flow; shear is 2D in four z layers. No solid/free-surface qualification.

LBM uses unchanged D2Q9 BGK production functions and upstream float32 weights.

Direct comparison is n32 shear only: same continuum parameters, actual face/node reference sampling.

Run duration includes diagnostics and raw sampling; this is not a repeated performance benchmark.

Matched advected shear: $t=0.5$ s, $\nu=0.1$



Matched transport phase and attenuation

